CAGE-Fake: Article Trustworthiness Audit Report

Calibration-Aware Gating and Evidence-Grounded Reasoning for Trustworthy Fake News Triage

LEGAL & METHODOLOGICAL DISCLAIMER: *A trained fake-news classifier predicts linguistic, stylometric, and semantic patterns learned from benchmark datasets; it does **NOT** independently verify whether a news story is factually true. URL extraction, model pattern recognition, and external fact verification remain strictly separate processes. High uncertainty cases are flagged for professional human editorial review.*

Source / URL: Manual Text Submission
Title: WHO Confirms New Global Health Preparedness Guidelines
Word Count: 37 words | **Date:** 2026-10-19 11:02:52

VERDICT: UNCERTAIN

Calibrated Confidence: 59.9%
Model Disagreement (std): 0.323
Gating Trigger: Low Confidence

1. Multi-Model Disagreement Analysis (5 Models)

Model Name	P(FAKE)	Model Vote	Confidence
Random Forest	71.4%	FAKE	71.4%
XGBoost	96.3%	FAKE	96.3%
LightGBM	97.2%	FAKE	97.2%
Logistic Regression	17.6%	REAL	82.4%
CAGE-Fake Stack	59.9%	FAKE	59.9%

Consensus Status: High Dissent / Conflicting Predictions
Inter-Model Standard Deviation: 0.2922 (Threshold Delta = 0.20)
Vote Distribution: 4 FAKE vs 1 REAL
Diagnostic: Inter-model disagreement (0.292) exceeds threshold (0.200).

2. News Credibility Fingerprint (Linguistic & Stylistic Scorecard)

Credibility Dimension	Score (0-100)	Journalistic Assessment
Sensationalism Discipline	100.0/100	Excellent
Source Attribution	0.0/100	Deceptive / Alert
Epistemic Grounding	100.0/100	Excellent
Punctuation Discipline	100.0/100	Excellent
Readability Structure	100.0/100	Excellent
Lexical Diversity	70.0/100	Adequate

Overall Credibility Index: 72.0/100
Credibility Tier: B (Standard News)

Scores are normalized against verified newsroom style guides: high attribution, balanced syntax, and low sensationalism maximize the index.

3. Transparent Reasoning & Counterfactual Sensitivity

Diagnostic Explanation:
1. Decision & Triage Status: The CAGE-Fake pipeline reached a verdict of **UNCERTAIN** with **59.9% confidence**.
- **Gating Trigger (Inter-Model Conflict)**: The ensemble exhibited significant dissent across the 5 models (std=0.292). Because models disagree, the system abstains from a definitive label to protect against false positives.

Counterfactual Simulation: Neutralizing sensational language and punctuation reduced FAKE confidence by 0.5% (from 59.9% to 59.3%), though underlying semantic patterns still lean deceptive.
Sensitivity Delta: P(FAKE) changed from 59.9% to 59.4% (-0.5%) when neutralizing sensational tokens.

4. Local Linguistic Signal Attributions

Top FAKE-Driving Indicators	Weight / Impact	Top REAL-Driving Anchors	Weight / Impact
p_xgb	+6.829	p_rf	-4.081
feature_10	+3.422	feature_5	-3.790

feature_11	+1.718	p_lgbm	-3.185
disagreement	+0.846	feature_8	-2.080

5. Empirical Benchmark Model Comparison (Held-out Test Split)

Model	Accuracy	Macro-F1	ECE	Coverage	Selective Acc
Random Forest	0.8639	0.8635	0.1548	100.0%	0.8639
XGBoost	0.8770	0.8769	0.0380	100.0%	0.8770
LightGBM	0.8689	0.8688	0.0418	100.0%	0.8689
Logistic Regression	0.8967	0.8967	0.0381	100.0%	0.8967
Uncalibrated Stack	0.8984	0.8983	0.0330	100.0%	0.8984
CAGE-Fake (Calibrated + Gated)	**0.9000**	**0.8999**	**0.0362**	**84.4%**	**0.9476**

6. Research Calibration & Disagreement Visualizations







